

Yang "William" Xu

Email | Web | GitHub | LinkedIn

Goal: Seeking a full-time position in machine learning research.

SKILLS

- **Research Topics**: Reinforcement Learning, LLM Evaluations, GenAI Modeling and Optimization
- **Programming Languages and Frameworks**: Python, C/C++, MATLAB, R, L^AT_EX, Linux, Git

EDUCATION

Purdue University , West Lafayette, USA	Aug 2022–Dec 2026
Ph.D. in Industrial Engineering (Operations Research Track).	
Purdue University , West Lafayette, USA	Aug 2022–May 2025
M.S. in Electrical and Computer Engineering	
The Chinese University of Hong Kong , Shenzhen, China	Sept 2018–May 2022
B.E. in Electronic and Information Engineering	

EXPERIENCE

- **Amazon.com Services LLC** May 2026–Present
Applied Scientist Intern Seattle, USA
 - Conduct research on LLM-based guided generation and reward modeling for improving business outcomes.
- **Purdue University** August 2022–Present
Graduate Research and Teaching Assistant West Lafayette, USA
 - Conduct research in machine learning on the theory and applications of decision-making and performance evaluation in stochastic systems, with publications at NeurIPS, ICML, AAAI, UAI, and TMLR.
 - Lead recitation sessions, hold in-person and online office hours, and grade exams in Math and IE departments.

PROJECTS

- **Decision-Making** August 2022–Present
 - Conducted research on reinforcement learning and stochastic approximation for decision-making, including robust, constrained, and quantum average-reward problems, with provable finite-sample guarantees.
 - Developed learning-based robotic navigation algorithms for dense crowds, emphasizing safe operation and strong generalization in multi-agent environments.
- **Performance Evaluation** August 2022–Present
 - Developed theory for policy evaluation in robust average-reward reinforcement learning and for Markov chain Poisson equations, targeting principled long-run performance analysis.
 - Contributed to [RLLTE](#) (470+ stars), a modular RL toolkit; and [SIREN](#), a reliable and statistically principled evaluation method of large language models.

RECENT PUBLICATIONS AND PREPRINTS

- **Y. Xu***, J. Zhang*, H. Sun, Z. Zhou, T. Cao, V. Aggarwal, "Towards Reliable LLM Evaluation: Correcting the Winner's Curse in Adaptive Benchmarking," <https://arxiv.org/abs/2605.05973>, 2026.
- J. Zhang, **Y. Xu**, V. Aggarwal, "Don't Freeze, Don't Crash: Extending the Safe Operating Range of Neural Navigation in Dense Crowds," <https://arxiv.org/abs/2603.06729>, 2026.
- **Y. Xu**, V. Aggarwal, "Persistent-Transient Policy Evaluation for Markov Chains via Minimal Peripheral Quotients," <https://arxiv.org/abs/2602.00474>, 2026.
- **Y. Xu**, S. Ganesh, V. Aggarwal, "Efficient Q -Learning and Actor-Critic Methods for Robust Average Reward Reinforcement Learning," *UAI*, 2026.
- **Y. Xu**, W. Mondal, V. Aggarwal, "Finite-Sample Analysis of Policy Evaluation for Robust Average Reward Reinforcement Learning," *NeurIPS*, 2025.
- **Y. Xu***, S. Ganesh*, W. Mondal, Q. Bai, V. Aggarwal, "Global Convergence for Average Reward Constrained MDPs with Primal-Dual Natural Actor Critic Algorithm," *NeurIPS*, 2025.
- B. Ganguly*, **Y. Xu***, V. Aggarwal, "Quantum Speedups in Regret Analysis of Infinite Horizon Average-Reward Markov Decision Processes," *ICML*, 2025.
- **Y. Xu**, V. Aggarwal, "Accelerating Quantum Reinforcement Learning with a Quantum Natural Policy Gradient Based Approach," *ICML*, 2025.
- M. Yuan, Z. Zhang, **Y. Xu**, S. Luo, B. Li, X. Jin, W. Zeng, "RLLTE: Long-Term Evolution Project of Reinforcement Learning," *AAAI*, 2025.

AWARDS

- NeurIPS Financial Aid Award 2025
- Dean's List Award, The Chinese University of Hong Kong, Shenzhen 2020, 2021